



Naba Husain

Electronics & Communication Engineering Student

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 github.com/naba-husain Indian naba-portfolio-psi.vercel.app

Education

2023 – 2027 Hubballi, India	Bachelor of Engineering (B.E.) in Electronics and Communication Engineering KLE Technological University CGPA : 8.3
2020 – 2022 Hubballi, India	Pre University in Science Chetan PU College
2020 Hubballi, India	Class X Smt. Nirmala K Thakkar Eng Med School

Skills

Programming Languages

- Python
- Java (Learning)
- SQL (Basic)

Computer Vision

- OpenCV
- Image Processing
- Object Detection
- Image Classification
- Helmet Detection

Java Programming

- Object-Oriented Programming
- Data Structures & Algorithms

Machine Learning

- Data Preprocessing
- Feature Engineering
- Model Evaluation
- Supervised Learning
- scikit-learn

Python Libraries

- Pandas
- NumPy
- Matplotlib

Generative AI

- Retrieval-Augmented Generation (RAG)
- Large Language Models (LLMs)
- Prompt Engineering
- Ollama
- AI Vector Search

Tools & Technologies

- Git
- GitHub
- VS Code
- MATLAB
- Jupyter Notebook

Languages

English	● ● ● ● ●	Hindi	● ● ● ● ●
Kannada	● ● ● ● ●	Urdu	● ● ● ● ●

Certificates

AI Foundations Associate
Oracle

Junos, Associate (JNCIA-Junos)
HPE Juniper

Generative AI
Oracle

Data Science
Oracle

MATLAB Onramp
MathWorks

Signal Processing Onramp
MathWorks

Projects

RAG-Based Video Question Answering with Timestamp Localization

Python | RAG | Ollama | NLP | LLM

- Developed a RAG-based system to answer questions from video content.
- Implemented timestamp localization to identify relevant video segments.
- Research paper accepted at INCECT 2026 based on this project.

AI-Powered HR Grievance Management System

Python | NLP | RAG | Generative AI

- Developed an AI-powered system to automate employee grievance management.
- Integrated NLP and RAG for intelligent grievance classification and query resolution.
- Designed a user-friendly platform for grievance submission and status tracking.

Earthquake-Stabilized Dialysis System for Patient Safety

Python | Machine Learning | Time-Series Analysis | Disaster Management

- Developed the Machine Learning module for the Smart India Hackathon project.
- Predicted earthquake-induced risks using real-time vibration data.
- Enabled AI-assisted safety decisions for dialysis machines during seismic events.

Physics-Aware Dataset Creation for Warehouse Robotics Deadlock Detection

ROS2 | Gazebo | Python | Autonomous Mobile Robots (AMRs)

- Developed a physics-aware dataset generation framework for warehouse robotics.
- Simulated deadlock scenarios using ROS2 and Gazebo for autonomous mobile robots.
- Generated labeled datasets to support deadlock detection and avoidance.

Helmet Detection System

Python | OpenCV | Computer Vision | Machine Learning

- Developed a computer vision system to detect helmet usage in real time.
- Applied image processing and object detection techniques using OpenCV.
- Improved detection accuracy through image preprocessing and model optimization.

Smart Accident Detection and Alert System

Arduino | Embedded Systems | Sensors | Bluetooth

- Designed a smart accident detection system using an accelerometer and Arduino.
- Detected sudden impacts and triggered instant alerts via Bluetooth.
- Improved emergency response through automated accident detection.

Awards

First Place – Technical Poster Presentation

INSIGNIA, SDM College of Engineering & Technology

Presented an engineering project and secured first place among participating teams.

First Place – PCB Design Competition

Department of Electronics & Communication Engineering, SDMCET

Secured first place in a PCB design competition demonstrating circuit design and implementation skills.

Smart India Hackathon 2025 – Internal Round Selection

KLE Technological University

Selected as Team ResQ in the Internal Smart India Hackathon 2025 for developing an AI-driven earthquake-stabilized dialysis system.

Publications

RAG-Based Video Question Answering with Timestamp Localization

International Conference on Emerging Computing Technologies (INCECT 2026)

- Accepted for presentation at INCECT 2026.
- Proposed a RAG-based framework for video question answering with timestamp localization.
- Leveraged LLMs to improve contextual video understanding.